STUDENT PERFORMANCE ANALYTICS SYSTEM

Python Project Documentation

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Course	EWB Courses
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Project	Student Performance Analytics System
Technologies	Python • NumPy • Pandas • Matplotlib

A practical Python data-analysis project for processing and visualizing student performance.

Project documentation prepared according to the EWB Courses assignment requirements.

1. Project Objective

The Student Performance Analytics System reads student records from a CSV file and analyzes academic performance using Python, NumPy, Pandas and Matplotlib. It calculates totals and averages, assigns grades, identifies pass/fail status, performs subject-wise analysis and finds top performers.

2. Project Features

Feature	Implementation
Data loading	Student records are loaded from students.csv using Pandas.
Total marks	The four subject marks are added for every student.
Average marks	The mean of the four subject marks is calculated.
Grades	A+, A, B, C, D and F are assigned using defined average ranges.
Pass/Fail	A student passes when every subject mark is at least 40.
Class analysis	Overall, highest and lowest averages plus pass/fail counts are calculated.
Subject analysis	Average marks are calculated for Python, NumPy, Pandas and AI.
Top performers	The three students with the highest averages are identified.
Visualization	Matplotlib creates three performance charts.

3. Technologies Used

Technology	Purpose
Python	Program logic, functions, conditions and processing.
NumPy	Numerical calculations such as mean, sum, maximum and minimum.
Pandas	CSV loading, DataFrame processing and statistical analysis.
Matplotlib	Charts and graphical presentation of results.
CSV	Stores the student dataset.

4. Dataset Description

The dataset contains 20 student records. Each record includes Student ID, Name, Department, marks in Python, NumPy, Pandas and AI, and Attendance.

ID	Name	Dept.	Python	NumPy	Pandas	AI	Attend.
101	Aarav	CSE	82	78	85	80	92
102	Priya	CSE	75	81	79	83	88
103	Rahul	CSE	91	89	94	92	95
104	Ananya	CSE	68	72	70	74	82
105	Vikash	CSE	55	59	57	61	76
106	Neha	CSE	88	92	90	89	94
107	Rohan	CSE	72	70	75	73	85
108	Sneha	CSE	95	94	96	97	98
109	Aman	CSE	61	65	63	68	79
110	Kavya	CSE	79	75	82	78	90
111	Arjun	CSE	84	86	88	85	93
112	Pooja	CSE	67	62	65	69	80

Table 1: Student dataset preview

5. Implementation Steps

Step	Description
1. Import libraries	Import NumPy, Pandas and Matplotlib.
2. Read CSV	Use pd.read_csv() to load students.csv.
3. Select subjects	Use a subject list so calculations can be reused.
4. Calculate total	Use NumPy/Pandas numerical operations to sum subject marks.
5. Calculate average	Calculate the mean of the four subject marks.
6. Assign grade	Apply the grading function to each student's average.
7. Determine result	Check whether every subject mark is at least 40.
8. Analyze data	Calculate class and subject-wise statistics.
9. Find top students	Sort/select the three highest averages.
10. Visualize	Generate charts using Matplotlib.
11. Save result	Write the final processed DataFrame to a CSV file.

6. Grading and Pass/Fail Rules

Average Marks	Grade
90 and above	A+
80–89	A
70–79	B
60–69	C
50–59	D
Below 50	F

Pass condition: A student is marked Pass when marks in all four subjects are at least 40. Otherwise, the result is Fail.

7. Final Analysis Results

Metric	Result
Number of students	20
Overall class average	78.58
Highest average	97.50
Lowest average	58.00
Passed students	20
Failed students	0
Best subject	AI
Lowest average subject	Python

8. Top 3 Performers

Rank	Student ID	Name	Average	Grade
1	120	Riya	97.50	A+
2	108	Sneha	95.50	A+
3	113	Aditya	92.50	A+

9. Subject-wise Performance

Subject	Average Marks
Python	77.25
NumPy	77.75
Pandas	79.05
AI	80.25

Observation: AI has the highest subject average (80.25), while Python has the lowest (77.25).

10. Visual Analysis

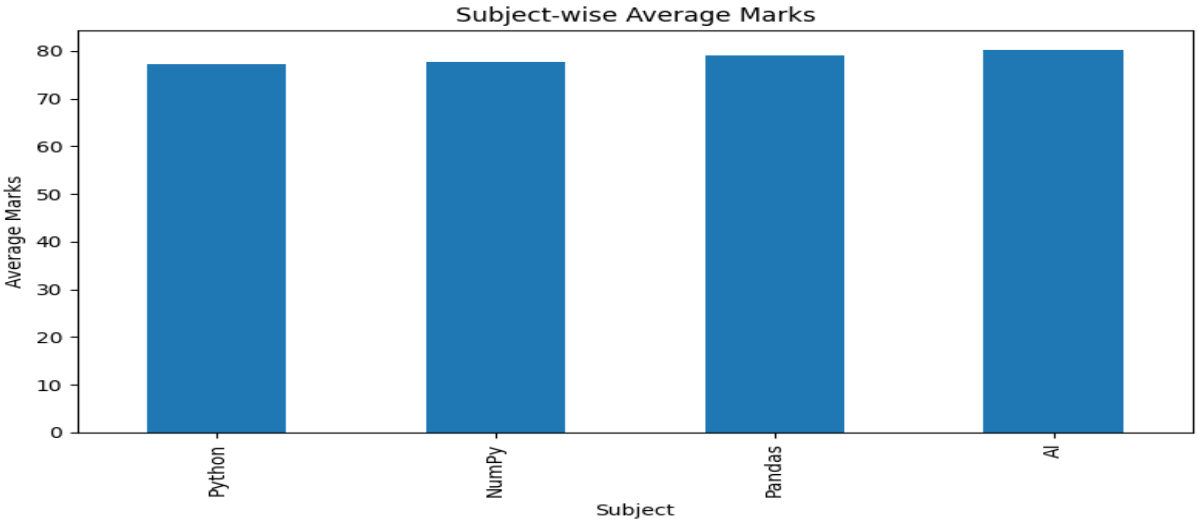


Figure 1: Subject-wise Average Marks

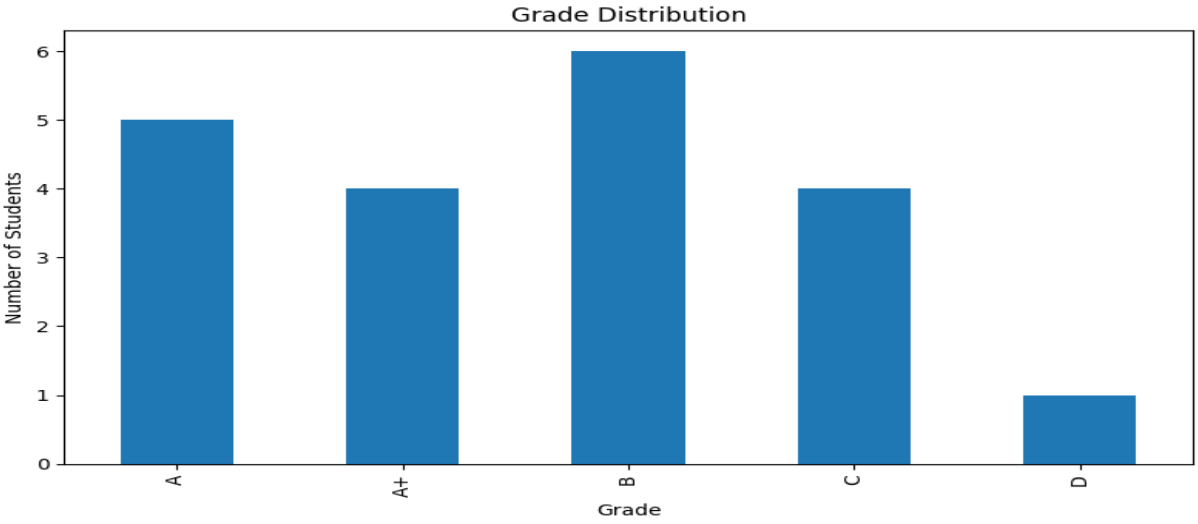


Figure 2: Grade Distribution

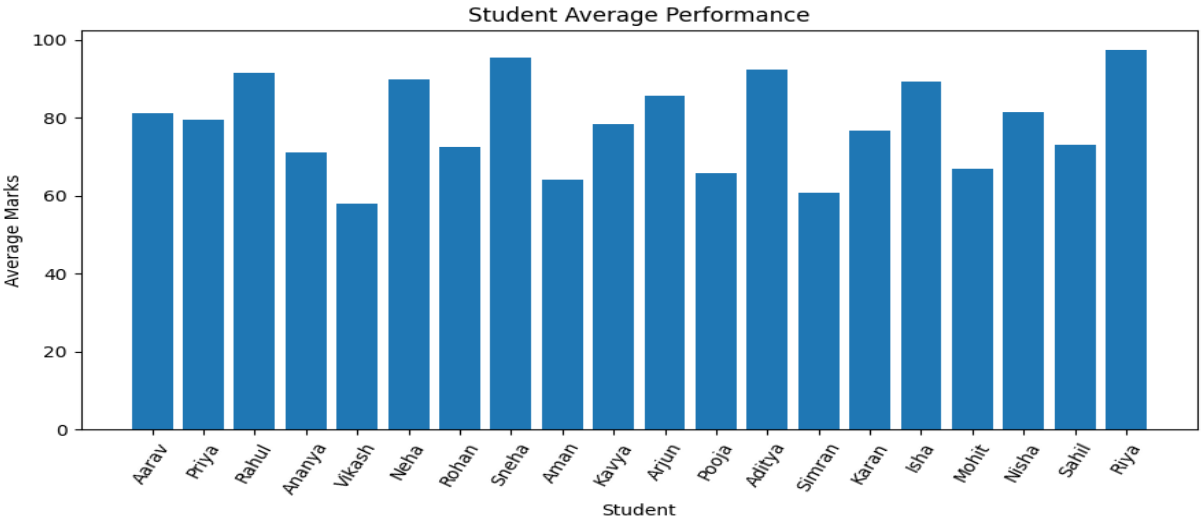


Figure 3: Student Average Performance

11. Final Program Output

The main results below are presented in a compact console-style format for easy verification.

```
============================================================
        STUDENT PERFORMANCE ANALYTICS SYSTEM
============================================================

[ CLASS SUMMARY ]
Students              : 20
Overall Class Average : 78.58
Highest Average       : 97.50
Lowest Average        : 58.00
Passed                : 20
Failed                : 0

[ SUBJECT-WISE AVERAGE ]
Python                : 77.25
NumPy                 : 77.75
Pandas                : 79.05
AI                    : 80.25
Best Subject          : AI
Lowest Avg. Subject   : Python

[ TOP 3 PERFORMERS ]
1. Riya       97.50  A+
2. Sneha      95.50  A+
3. Aditya     92.50  A+

[ GRADE DISTRIBUTION ]
A+ : 4
A  : 5
B  : 6
C  : 4
D  : 1

[ NUMPY ANALYSIS ]
Overall Marks Average : 78.58
Highest Subject Mark  : 99
Lowest Subject Mark   : 55
============================================================
```

Figure 4: Compact final program output

12. Detailed Student Performance Output

ID	Name	Total	Average	Grade	Result
101	Aarav	325	81.25	A	Pass
102	Priya	318	79.50	B	Pass
103	Rahul	366	91.50	A+	Pass
104	Ananya	284	71.00	B	Pass
105	Vikash	232	58.00	D	Pass
106	Neha	359	89.75	A	Pass
107	Rohan	290	72.50	B	Pass
108	Sneha	382	95.50	A+	Pass
109	Aman	257	64.25	C	Pass
110	Kavya	314	78.50	B	Pass
111	Arjun	343	85.75	A	Pass
112	Pooja	263	65.75	C	Pass
113	Aditya	370	92.50	A+	Pass
114	Simran	243	60.75	C	Pass
115	Karan	307	76.75	B	Pass
116	Isha	357	89.25	A	Pass
117	Mohit	268	67.00	C	Pass
118	Nisha	326	81.50	A	Pass
119	Sahil	292	73.00	B	Pass
120	Riya	390	97.50	A+	Pass

Table 2: Complete calculated output for all 20 students

13. Challenges and Learning

- Organizing student records in CSV format.
- Using Pandas DataFrames for structured processing.
- Applying the same calculations to all student records.
- Using NumPy for numerical analysis.
- Creating grading and pass/fail conditions.
- Comparing subject-wise performance.
- Creating readable charts using Matplotlib.
- Saving the processed analysis as a new CSV file.

14. Final Outcome

The Student Performance Analytics System successfully processes 20 student records and produces totals, averages, grades, pass/fail status, class statistics, subject-wise averages, top performers and visual charts. The project demonstrates a complete basic data-analysis workflow using Python, NumPy, Pandas and Matplotlib.

15. Conclusion

The project converts raw student data into useful academic information. It provides a simple and understandable way to evaluate individual and class performance and demonstrates practical use of Python data-analysis libraries.

16. GitHub Repository

GitHub Repository Link: To be added after uploading the project.

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